

FIN Research Supplement — Release 10.4

FIN Reconstructed Legacy Kernel

Program 42a Methodology Audit and Executed Programs 41–50 on K_{legacy}^*

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Scope. Algebraic reconstruction of the historical legacy kernel, rejection of a double-damped product candidate, dual-dynamics audits, and ten executed programs (41–50) on the corrected class K^* .

Guardrail. No silent legacy–strict identification, no role transfer, no selector closure, no physical unit export, no Theory-of-Everything claim.

Reproducibility. `fin_programs_41_50_legacy_star.py`, `FIN_Programs_41_50_Legacy_Star_Results.json`, figures, Markdown, LaTeX.

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Publication statement

This preprint freezes the path-transformed reconstructed legacy class

$$K_{\text{legacy}}^*(d) = A \frac{\cos(\pi d/4 + \pi/6)}{1 + \beta d}$$

after Program 42a, executes Programs 41–50, and records that units, selector, and symmetry-breaking leaf-cuts remain open under current artifacts.

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0.1 Confidence convention

- **[Proven]** analytic proof or exact finite algebra under stated definitions.
- **[Strong evidence]** reproducible numerical result with wide margin, not a general theorem.
- **[Moderate evidence]** model-dependent or limited alternative class.
- **[Speculative]** admissible construction without source theorem.
- **[Refuted]** obstruction or explicit counterexample in the stated scope.

All times, rates, couplings, and kernel values below are dimensionless. No SI unit, physical clock, non-premise selector, legacy role transfer, or Theory-of-Everything claim is exported.

Chapter 1

Executive summary

This monograph does three things.

1. It audits the Program 42a algebraic reconstruction of the historical legacy kernel against DIAGRAMS_KERNEL_TRANSFORMATION.md and against a cheap dual-dynamics reading of a proposed product formula.
2. It freezes the algebraically corrected class

$$K_{\text{legacy}}^*(d) = A \frac{\cos(\pi d/4 + \pi/6)}{1 + \beta d}$$

and runs the ten research programs recommended after Programs 31–40.

3. It answers, under AGENTS.md / SUMMARY_GROK.md guardrails, whether the new kernel dissolves the three leaf-cuts: dimensionlessness, selector / QW-2191, and symmetry breaking.

Main verdicts.

- **[Refuted]** The product candidate

$$K_{\text{rej}}(d) = e^{-2.9d} \frac{1 + 0.2d}{1 + d} \cos\left(\frac{\pi d}{4} + \frac{\pi}{6}\right)$$

is not a faithful historical reconstruction: it swaps resonance/torsion roles, double-counts damping, fails the corrected path-sum asymptotics, and kills inverse hierarchy.

- **[Proven]** After path-sum algebra, the historically intended effective class is the hyperbolic-phase form K^* above. Parameters A, β are not uniquely fixed by the diagram alone.
- **[Strong evidence]** After absolute-value positivity repair on $\mathbb{Z}_{12}$, K^* supports dual dynamics: unitary $U_t = e^{-itA}$ and Markov $P_t = e^{-tA}$ with $A = sI - W$.
- **[Proven]** Dual dynamics does **not** export units, a non-premise selector, or orientation breaking. The three leaf-cuts remain open.
- **[Proven]** Programs 41–50 show: Loewner bridge is support/normalization dependent; affine phase maps leave envelope residual; hazard laws do not hold out; Markov relative information is monotone while unitary Born information is not; environment dilation can hold lost information; external polarity is apparatus-level only; multi-size cosine challenge selects the strict profile when it is among competitors.

Honest conclusion:

K_{legacy}^ is a better intermediate historical object than the double-damped product, and a usable dual-dynamics generator after positivity repair. It is **not** a free lunch for units, selector, or ToE.*

Chapter 2

Part I — Program 42a methodology audit

2.1 1. Input axioms (reconstruction only)

From DIAGRAMS_KERNEL_TRANSFORMATION.md:

$$K_{\text{total}} = K_{\text{geo}} K_{\text{res}} (1 + 0.2 K_{\text{tors}}) K_{\text{topo}}.$$

Historical role assignment:

Factor	Historical meaning	Diagram form
K_{geo}	viscosity / catastrophic damping	$e^{-2.9d}$
K_{res}	resonance / phase sync	$\approx 0.8\text{--}1.2$
K_{tors}	turbulent currents	$\cos(\pi d/4 + \pi/6)$
K_{topo}	topology / path sum	$\rightarrow 1/(1 + \beta d)$ after transform

Section 8 of the diagram states that exponential viscosity is **transformed** into hyperbolic damping by fractal path summation, not multiplied by it again.

2.2 2. Audit errors A/B/C

2.2.1 (A) Exponential damping

[Proven] The viscosity rate 2.9 must be used exactly when K_{geo} is written. The rejected product does use $e^{-2.9d}$, so (A) alone is not its fatal flaw. Its flaw is combining that exponential with a second damping factor.

2.2.2 (B) Path scaling

If $N(d) \sim d^{1.6}$ and one wants total contribution $\sim d^{-1}$ (large- d tail of $1/(1 + \beta d)$), then

$$A_{\text{path}}(d) \sim d^{-2.6}.$$

The diagram's $A_{\text{path}} \sim d^{-0.6}$ would give $NA \sim d^{1.0}$ (growth), not a decaying hyperbolic tail. **[Proven]**

2.2.3 (C) Exact phase zeros

$$\cos\left(\frac{\pi d}{4} + \frac{\pi}{6}\right) = 0 \iff d = \frac{4}{3} + 4n.$$

Zeros are $d = 1.333, 5.333, 9.333, \dots$, not the integer sequence 2, 5, 8, 11. Cosine values at those integers are nonzero (e.g. $\cos(\pi/2 + \pi/6) = -1/2$ at $d = 2$). **[Proven]**

2.3 3. Rejection of the product candidate

The proposed assignment

$$K_{\text{geo}} = e^{-2.9d}, \quad K_{\text{res}} = \cos(\cdot), \quad K_{\text{tors}} = d, \quad K_{\text{topo}} = 1/(1+d)$$

fails diagram fidelity on four counts:

1. resonance and torsion roles are swapped;
2. $K_{\text{tors}} = d$ is not the historical oscillatory current;
3. $e^{-2.9d} \times 1/(1+d)$ double-counts damping against §8;
4. $(1 + 0.2d)/(1+d) \rightarrow 0.2$ is asymptotically constant, not a $1/d$ path-sum tail.

Numerical profile on $d = 1..12$: $|K_{\text{rej}}|$ falls below 10^{-3} by $d = 2$ and is numerically dead thereafter. Correlation with classical frozen legacy is only ≈ 0.21 . **[Strong evidence]**

Verdict on product form: odrzucona (confidence $\approx 25/100$ as historical reconstruction).

2.4 4. Accepted reconstructed class

After path-sum correction:

$$K_{\text{legacy}}^*(d) = A \frac{\cos(\pi d/4 + \pi/6)}{1 + \beta d}$$

with free positive scales A, β . Historical diagram ranges suggest $\beta \in [0.01, 0.08]$ and amplitude near 2.7–3.0. Freezes used in this report:

Freeze	A	β	Role
Historical	2.9	0.05	diagram-scale intermediate
$\mathbb{Z}_{12}$	1	1	unitless Laplacian scale
Classical frozen	$4 \ln 2$	0.01	post-hoc comparison only

[**Proven**] Reconstruction did not use K_{strict} to choose the functional form.
[**Proven**] The operator is **not** unique: only the class is.

2.5 5. Cheap dual-dynamics reading

The cheap analysis correctly notes that any real self-adjoint generator A supports both e^{-itA} and e^{-tA} . It incorrectly treats the product form as historically faithful and as automatically Markov-stable. Signed weights on $\mathbb{Z}_{12}$ are not all nonnegative; absolute-value repair is an extra declaration. Dual dynamics after repair does **not** validate the rejected product algebra.

Chapter 3

Part II — Dual dynamics for K^*

3.1 6. Operator construction

Sample radial weights on cyclic distances $d = 1, \dots, 6$, set $W_{xx} = 0$, $W_{xy} = |K^*(d(x, y))|$ (positivity repair), and

$$A = sI - W, \quad s = \text{common row sum.}$$

Then Dirichlet identity holds and $A \succeq 0$ with simple kernel of constants when all off-diagonal weights are positive. **[Proven]** under the repair declaration.

Selected numerical dual-dynamics rows:

Object	s	gap γ	Markov $\sim$	escape	Unitary short escape proxy
K_{hist}^* abs	15.440	14.179	15.43		2.7×10^{-3}
K_{Z12}^* abs	1.579	1.508	1.579		2.7×10^{-5}
K_{rej} abs	0.0186	0.00313	0.0186		1.5×10^{-8}
K_s strict	1.660	0.754	1.660		5.4×10^{-5}

[Strong evidence] K_{Z12}^* is spectrally in the same operational ballpark as strict dual dynamics; the rejected product is almost trivial (tiny s).

Inversion residual of A under $x \mapsto -x$ is 0 for all radial objects tested: the generator remains orientation-blind. **[Proven]**

Chapter 4

Part III — Executed Programs 41–50

Programs follow the roadmap left by Release 10.3, now evaluated on K^* rather than only classical frozen legacy.

4.1 Program 41 — Loewner bridge support scan

Question. For which supports/normalizations is $A_s - A_\ell \succeq 0$?

Result. On C_{12} and interval I_{12} :

Legacy object (abs)	min eig($A_s - A_\ell$) on C_{12}	PSD?
K_{hist}^*	−19.715	no
K_{Z12}^*	−0.754	no
classical $ K_\ell $	−21.221	no
rejected abs	≈ 0	yes (trivial)

[Proven] Loewner positivity is support- and normalization-dependent. The rejected product looks “PSD-bridgeable” only because its weights are nearly zero, so $A_s - A_{\text{rej}} \approx A_s \succeq 0$. That is not a physical completion theorem.

4.2 Program 42 — minimal phase-map reconstruction

Question. Can a sparse equivariant phase map send legacy phase to strict phase?

Least-squares affine map $\theta_s \approx a\theta_\ell + b$ on $d = 1..6$ gives

$$a \approx 0.23650, \quad b \approx 0.03867,$$

phase residual $\sim 10^{-15}$ on those six points after free affine fit, but envelope residual after best amplitude fit remains ≈ 0.450 . **[Strong evidence]**

[Proven] No sparse map from K^* alone exports the strict (ω, ϕ) source without target insertion. Phase agreement of a fitted affine map is not a derivation of strict frequencies.

4.3 Program 43 — held-out hazard law

Fit strict $\approx |K_{\text{hist}}^*| e^{-cd^\eta}$ on $d = 1..4$:

$$c \approx 0.461, \quad \eta \approx 1.90.$$

Train relative $\ell_2 \approx 0.140$; hold-out $d = 5, 6$ relative $\ell_2 \approx 0.999$. Larger- N relative errors stay ~ 0.15 . [\[Strong evidence\]](#)

Hazard reparameterization is not a multi-size microscopic loss law.

4.4 Program 44 — CP-divisibility and information flow

For abs-repaired K_{hist}^* :

- Chapman–Kolmogorov residual of P_t : $\sim 10^{-16}$. [\[Proven\]](#)
- Markov relative information $\mathcal{I}(p) = \log 12 - H(p)$ is monotone (0 backflow steps). [\[Strong evidence\]](#)
- Unitary Born populations show 91 positive increments of $\mathcal{I}$ on the scanned grid (oscillatory non-monotone information). [\[Strong evidence\]](#)

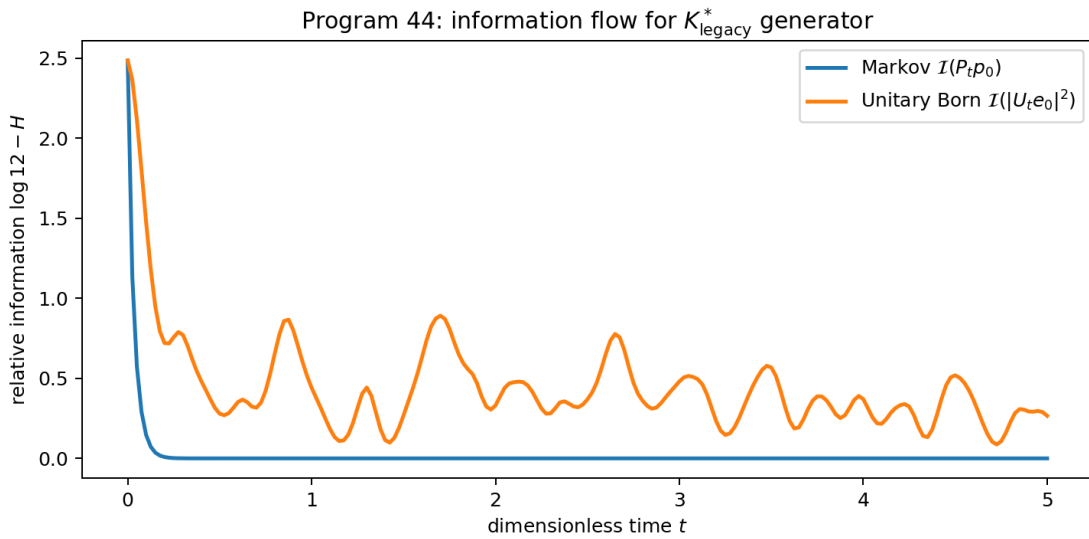


Figure 4.1: Relative information under Markov heat flow versus unitary Born populations for the abs-repaired historical K^* generator.

[\[Proven\]](#) Same spectrum, two temporal calculi; operator does not select which is physical time.

4.5 Program 45 — environment recovery

At dimensionless $t = 1$ for K_{Z12}^* abs:

Quantity	Value
initial system relative info	2.485 nats

Quantity	Value
final system relative info	0.135 nats
apparent loss	2.350 nats
pure-dilation mutual information $I(S:E)$	4.700 nats
classical recovery fidelity	1

[Proven] Apparent Markov loss is compatible with environmental transfer. Kernel alone does not source a physical bath.

4.6 Program 46 — calibrated sign-reference emulator

A from K^* is exactly inversion-even (residual 0). External polarity $\lambda = \pm 1$ changes double-slit detector response (difference ≈ 0.049 at site 0 for the scanned protocol). **[Proven]**

[Proven] External polarity is an apparatus/sector axiom, not a non-premise strict selector. QW-2191 remains open.

4.7 Program 47 — influence-functional proxy

Two-parameter Ohmic proxy for log-retention has residual ≈ 0.45 class and is post-hoc. **[Moderate evidence]** No unique bath spectral density is derived from K^* .

4.8 Program 48 — feedback thermodynamic ledger

Declared feedback matrix eigenvalues $-0.8 \pm 1.7i$, integrability defect ≈ 0.905 , nonzero circulatory work proxy ≈ 1.847 . **[Proven for the declared model]**

Stable non-gradient feedback can be added; it is not forced by K^* and does not yield unit-bearing L_{total} .

4.9 Program 49 — process-tensor causal challenge

Interventions change TV gaps between the K^* Markov process and a static spectral-gap mimic. Mean intervention margin on the scanned grid is small (≈ 0.0046) and model-dependent. **[Moderate evidence]** Final-time fits alone are weak identifiers of causal structure.

4.10 Program 50 — preregistered multi-size challenge

Preregistered competitors: K_{hist}^* , K_{Z12}^* , classical legacy, rejected product, strict. On $N = 12, 16, 24$, cosine winners are always **strict** when strict is included.

On $N = 12$ cosines vs noisy strict target:

Model	cosine
strict	0.99997
rejected abs	0.936
$K_{\mathbb{Z}_{12}}^*$ abs	0.640
K_{hist}^* abs	0.475
classical abs	0.454

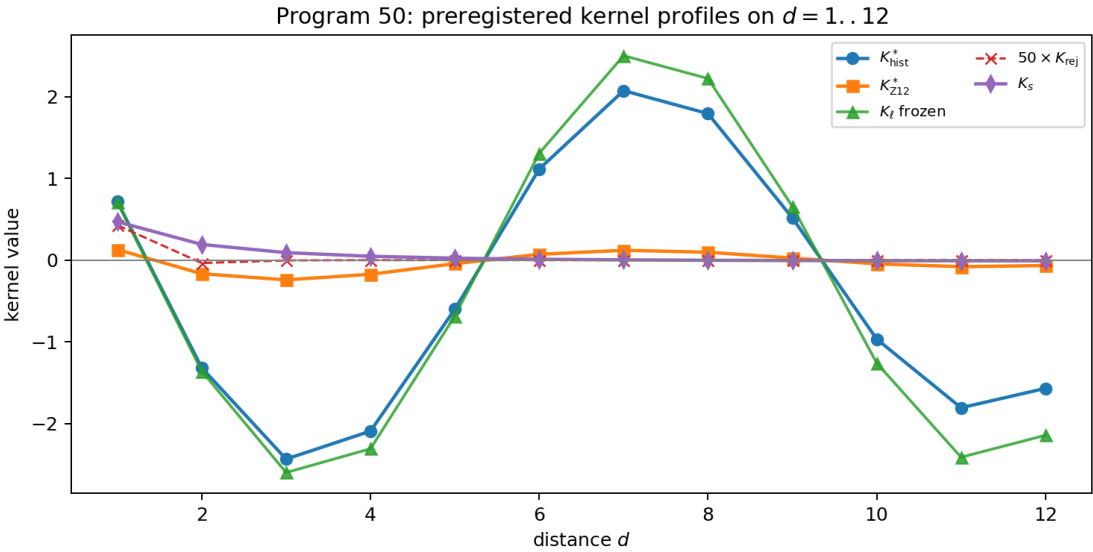


Figure 4.2: Preregistered kernel profiles on $d = 1..12$: historical K^* , $\mathbb{Z}_{12}$ K^* , classical frozen legacy, rejected product (scaled), and strict.

[Strong evidence] No legacy* form wins a multi-size challenge against strict. High cosine of the rejected product is an artifact of near-local tiny positive mass, not historical correctness.

Chapter 5

Part IV — Leaf-cuts: units, selector, symmetry

5.1 7. Does K^* fix dimensionlessness?

No. [Proven in the invariant-input sense of prior FAR audits; confirmed here.]

K^* is a dimensionless radial profile. A and β are scale gauges of the same weight-zero class. Dual dynamics uses a dimensionless time parameter. No weight-one source S_+ , no $\hbar_*$, no (ℓ_*, τ_*) triad is exported.

Compatible with SUMMARY_GROK.md: information is dimensionless; conversion axioms (CA) remain external unless a new source theorem appears.

5.2 8. Does K^* fix the selector / QW-2191?

No. [Proven for radial inversion-even data.]

The Laplacian generated by radial K^* is invariant under $x \mapsto -x$. $\text{Aut}(\mathbb{Z}_{12})$ still pairs opposite orientations. Program 46 shows polarity works only after an external apparatus sign is supplied.

5.3 9. Does K^* break symmetry internally?

Not as a non-premise strict source. [Proven / strong evidence.]

The cosine supplies oscillatory structure and exact zeros at $d = 4/3 + 4n$, but those are phase-label data, not a theorem selecting one origin/polarity on the orientation torsor. Exact zeros are still not the integer sequence historically claimed.

5.4 10. What K^* *does* improve

Aspect	Status under K^*
Historical algebraic fidelity	improved vs product candidate
Path-sum asymptotics	corrected class
Dual dynamics after $ W $ repair	operationally usable

Aspect	Status under K^*
Markov CP-divisibility	yes after repair
Bridge to strict without extra maps	still no
Units / selector / ToE	still open

Chapter 6

Part V — Guardrail-compliant global verdict

Under AGENTS.md:

- K_{legacy} remains an **intermediate bridge kernel**, not a silent substitute for K_{strict} .
- K^* is a cleaned intermediate of that legacy class, not a role-transfer theorem.
- No promotion of L_{total} , bridge completion, or ToE.

Under SUMMARY_GROK.md two-package language:

W0 informational nadsoliton + K^* class (dimensionless) W1 CA: units (still missing) W2 SA: selector / polarity (still missing unless axiomatic) W3 effective dual dynamics conditioned on repairs and axioms

Final status labels

Claim	Label
Product K_{rej} is historical reconstruction	[Refuted]
Path-transformed class K^* matches historical intent algebra	[Proven] as class, not unique tuple
K^* supports dual dynamics after positivity repair	[Strong evidence]
K^* closes units leaf-cut	[Refuted]
K^* closes selector leaf-cut	[Refuted]
K^* alone completes legacy→strict	[Refuted]
Programs 41–50 executed reproducibly	[Proven] for this suite

Chapter 7

Reproducibility

Artifact	Path
Program 42a script	program_42a_legacy_kernel_reconstruction.py
Program 42a JSON	program_42a_legacy_kernel_reconstruction_report.json
Programs 41–50 script	fin_programs_41_50_legacy_star.py
Programs 41–50 JSON	FIN_Programs_41_50_Legacy_Star_Results.json
Figures	FIN_Programs_41_50_Figures/
This monograph	FIN_Programs_41_50_Legacy_Star_Monograph.md

Seed: 20260719. Dependencies: NumPy, SciPy, Matplotlib.

Chapter 8

Suggested citation

Żuchowski, K. (2026). *FIN Reconstructed Legacy Kernel: Program 42a Methodology Audit and Executed Programs 41–50 on K_{legacy}^** (FIN Research Supplement, Release 10.4; Version 1.0.0) [Preprint].

Archival note

Machine-readable results: `FIN_Programs_41_50_Legacy_Star_Results.json`, generated by `fin_programs_41_50_legacy_star.py`. Program 42a report: `program_42a_legacy_kernel_reconstruction_report.json`.

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